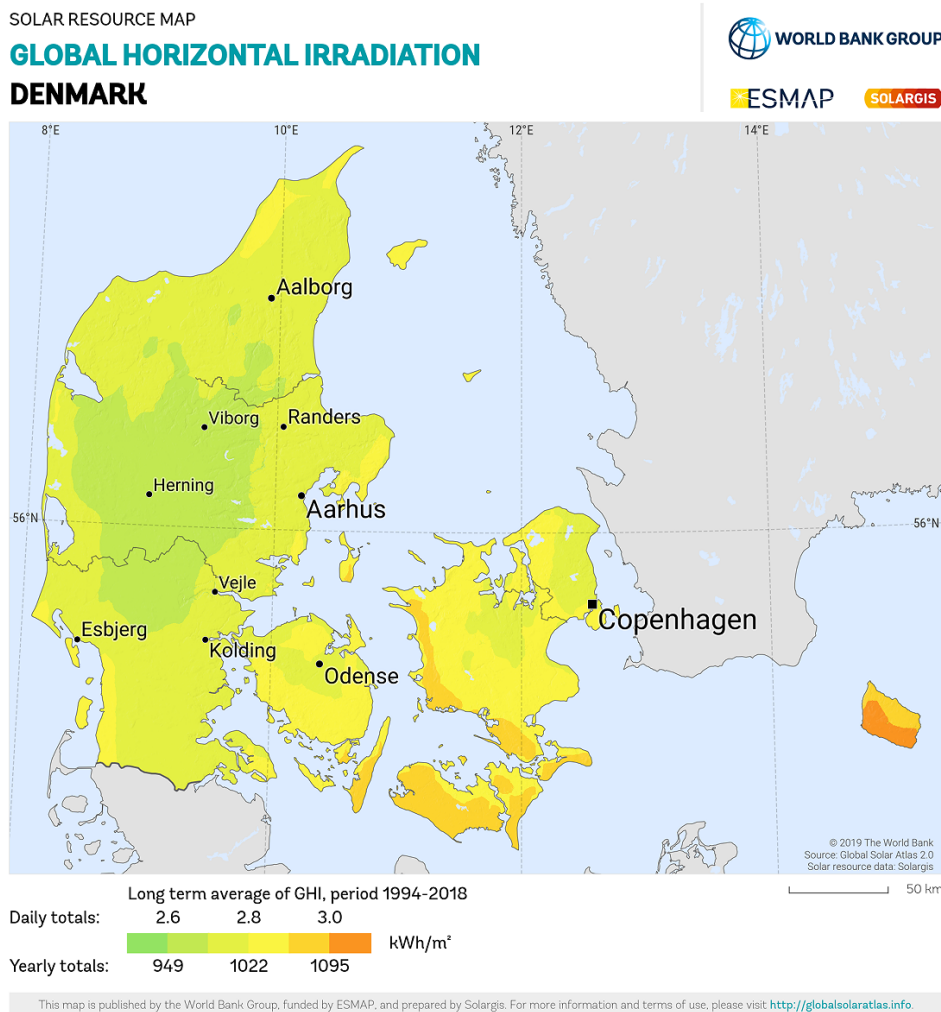


Warm-up problem 3 Modelling 2025

3. Assuming a forest somewhere in Denmark has a net primary productivity of $5 \text{ Mg ha}^{-1} \text{ yr}^{-1}$ on a carbon (C) basis, estimate the net efficiency of solar energy capture.

$$\eta_{\text{NPP}} = \frac{\dot{E}_{\text{NPP}}}{\dot{E}_{\text{sol}}}$$

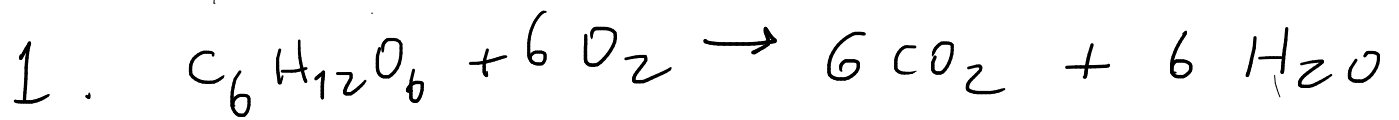
Let's evaluate this on an annual basis,
so we need average annual solar
radiation and the energy content
of biomass.



$\sim 1000 \text{ kWh/m}^2$
'yr

<https://solargis.com/resources/free-maps-and-gis-data?locality=denmark>

2 different approaches for biomass energy.



$$\Delta H \approx 2.8 \text{ MJ/mol}$$

$$= 2.8 \cdot 10^6 \text{ J} / 180 \text{ g}$$

$$\approx 16 \text{ kJ/g}$$

2. LHV of biomass $\sim 10^{-18} \text{ GJ/t}$

$$10^{-18} 10^9 \text{ J} / 10^6 \text{ g}$$

$$= 10^{-18} \text{ kJ/g}$$

Assume biomass is $\sim 50\% \text{ C}$

$$\text{and } 5 \text{ Mg C/ha-yr} = 5 \cdot 10^6 \text{ g C} / 10^4 \text{ m}^2 \cdot \text{yr}$$

$$= 500 \text{ g C/m}^2 \cdot \text{yr} \approx 1 \text{ kg biomass/m}^2 \cdot \text{yr}$$

$$1000 \text{ g biomass} / \text{m}^2 \cdot \text{yr} \cdot 16 \text{ kJ} / \text{g biomass}$$

$$= 16 \text{ MJ} / \text{m}^2 \cdot \text{yr}$$

$$\text{Solar energy} : 1000 \text{ kWh} / \text{m}^2 \cdot \text{yr}$$

$$= 1000 \cdot 10^3 \text{ J/s} \cdot 3600 \text{ s} / \text{m}^2 \cdot \text{yr}$$

$$= 3.6 \text{ GJ} / \text{m}^2 \cdot \text{yr} = 3600 \text{ MJ} / \text{m}^2 \cdot \text{yr}$$

$$\eta_{\text{NPP}} = \frac{16}{3600} = 0.44 \%$$